Classification of Self-Actions and the Unique Involutive Survivor

Abstract

We classify self-actions $a : \alpha \to \alpha$ by local orbit shape. Rest, asymmetric step, and erasure each meet a formal obstruction in the ambient of pointed symmetric steps; the surviving shape is the fixed-point-free two-cycle. That survivor is characterised by a universal property: $(\mathsf{Bool}, \neg)$ is initial among pointed unoriented acts, and every fundamental act reads as negation. From fundamentality we obtain a Lawvere-style seal (no point-surjective self-reading), an ω-ladder of namers, and—after transferring losslessness to forward microsteps by two-bounce factorisation—registration, capacity, and a single $\mathbb{Z}/2\mathbb{Z}$-underdetermination with three faces. Canonicity of the two-bounce pair fails; channel *swap* nevertheless covers step reversal. Content gauges are separated from packaging gauges so that factorisation non-uniqueness does not inflate the dictionary $\mathbb{Z}/2\mathbb{Z}$. The development is machine-checked in Lean 4 (no Mathlib, no `sorry`). Physical readings appear only in a dictionary layer of certificates.

Keywords. classification; involution; universal property; Lawvere diagonal; underdetermination; gauge; machine-checked mathematics.

1 Introduction

The central contribution of this paper is an *eliminative classification of self-actions*. Given $a : \alpha \to \alpha$, the local possibilities at a state are rest, two-cycle, asymmetric step, and—as a pairwise defect—erasure. In a precisely specified ambient, every shape except one encounters an obstruction. The survivor is the live two-cycle. Negation on Bool [3] is not assumed at the outset; it is the unique object (up to label swap) satisfying the universal property, in the standard categorical sense [14], of that survivor.

The formal development then follows a fixed dependency:

$$\begin{tikzcd}
\text{classification} \arrow[r] & \text{fundamentality} \arrow[r] & \text{seal} \arrow[r] & \text{ladder} \arrow[d] & & & & \\
& & & \text{registration} \arrow[r] & \text{archive} \arrow[r] & \text{capacity} \arrow[r] & \text{arrow} \arrow[r] & \text{interpretations}
\end{tikzcd}$$

Each arrow is a theorem or a defined interface; philosophical commentary is postponed until the supporting machinery is in place.

Method and layers. The text maintains three layers.

(1) **Formal mathematics** — definitions, theorems, proofs.

(2) **Philosophical interpretation** — what the mathematics suggests, marked as such.

(3) **Dictionary** — optional physical or contemplative identifications, admitted only by certificate.

Claims are tagged by status: proved, corollary, interpretation, open, refused, or philosophical residue. The companion Lean 4 corpus [7] (no Mathlib [15], no `sorry`) supplies declaration names; this paper is self-contained at the level of statements.

Roadmap. Section 2 gives the architecture. Sections 3–4 carry out the classification and the universal characterisation of the survivor. Sections 5–6 obtain seal and ladder. Sections 7–9 transfer losslessness, record registration and capacity, and identify the arrow with address. Section 10 states the single $\mathbb{Z}/2\mathbb{Z}$ and the gauge demarcation. Section 11 confines dictionary readings. Section 12 collects open, refused, and philosophical items. Section 13 situates the development in prior work. Section 14 concludes.

2 Architecture

The companion corpus is stratified into three layers with downward-only imports (Figure 1).

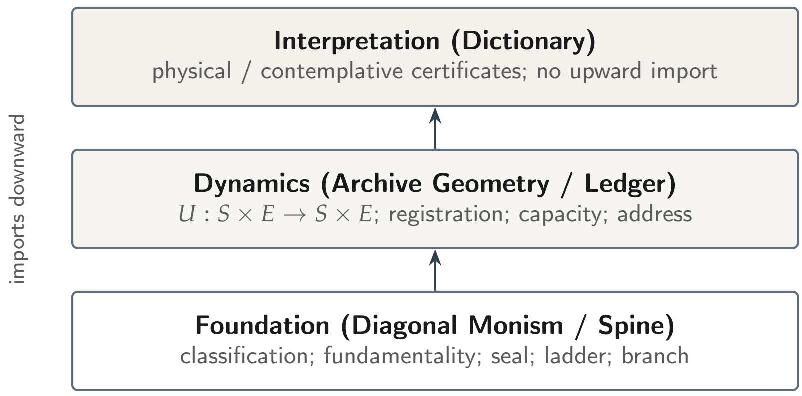


Figure 1: *Three-layer architecture. Arrows indicate permitted dependence (foundation supports dynamics; dynamics support interpretation). The figure orients the reader; it proves nothing.*

Within the foundation, the dependency is classification $\to$ fundamentality $\to$ seal $\to$ ladder. The dynamics layer imports losslessness along an interface (registration via two-bounce), realises archive and capacity on the ladder carrier, and exports address as the arrow reading. The dictionary layer consumes those exports; it does not feed them.

3 Classification of self-actions

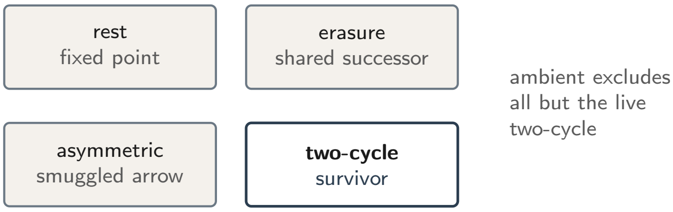
Definition 3.1 (Arena data)**.** Fix a type α and a map $a : \alpha \to \alpha$. Write

$$\mathsf{Step}(a)(x, y) \;:\Leftrightarrow\; y = a(x).$$

Call a a *symmetric step* when $\mathsf{Step}(a)$ is a symmetric relation; equivalently, $a \circ a = \mathrm{id}$. Call a *live* when $\forall x.\, a(x) \neq x$. Call a *lossless* when a is injective. Call a *erasing* when distinct states share a successor.

Theorem 3.2 (Pointwise classification)**.** *For decidable equality on α, at each z exactly one of the following holds: (i) rest ($a(z) = z$), with all iterates stagnant; (ii) two-cycle at z ($a(z) \neq z$*

and $a(a(z)) = z$), with every iterate equal to z or $a(z)$; (iii) asymmetric step at z ($a(a(z)) \neq z$). The three cases are pairwise incompatible. Erasure is diagnosed separately as a pairwise defect of losslessness.



Remark 3.3 (Status). Theorem 3.2 is proved. It is the eliminative engine of the paper: subsequent sections impose ambient constraints under which (i) and (iii), and erasure, are excluded as candidates for fundamentality.

Symmetric step is the ambient equation used below. It jointly excludes asymmetric arrow (case (iii) globally) and, by the next corollary, erasure.

Proposition 3.4 (In-arena losslessness)**.** *If a is a symmetric step, then a is not erasing.*

Remark 3.5 (Status). Corollary of Definition 3.1 (involution $\Rightarrow$ injectivity). Not an independent creed.

Interpretation 3.6. The choice of symmetric step as ambient is a modelling decision: it bans smuggled time (asymmetric step) and smuggled loss (erasure) by one equation. Whether that ambient is the "right" one for fundamentality is philosophical residue (Section 12), not a theorem.

4 Fundamentality as universal property

Definition 4.1 (Pointed unoriented acts)**.** Objects are triples (α, a, z) with a a symmetric step and $z : \alpha$ a seed. Morphisms are pointed equivariant maps.

Rest is incompatible with liveness. Asymmetric step is incompatible with symmetric step. Erasure is incompatible with Proposition 3.4. Relative to Definition 4.1 and liveness, the only remaining pointwise shape is the live two-cycle.

Theorem 4.2 (Canon / initiality)**.** *In the ambient of pointed unoriented acts, $(\mathsf{Bool}, \neg, \mathsf{true})$ is initial. An act is* fundamental *precisely when it is live, symmetric, and generated from a seed by the two-cycle reading; every fundamental reads as $\neg$; any two fundamentals correspond uniquely up to the pointed reading.*

Remark 4.3 (Status). Proved. Negation is the unique (up to label swap) object satisfying the universal property of the live two-cycle in this ambient—the survivor of the classification, not a primitive assumption.

Interpretation 4.4. "The fundamental" names that initial object after uniqueness. Excluded fixed point ("void") is liveness restated. Label swap of the two poles is a content gauge: exported readings mention it; see Section 10.

5 Seal

Dependence: *Theorem 4.2 (liveness of fundamentals).*

The engine is Lawvere's diagonal argument [13], the common abstraction of Cantor's theorem [5] and kindred fixed-point arguments [19].

Theorem 5.1 (Seal on a fundamental). *Let a be live. No map $f : A \to (A \to \mathsf{Bool})$ is point-surjective: the diagonal dodge $x \mapsto \neg(f\,x\,x)$ is missed. Specialising to fundamental acts, every self-reading admits an escape.*

Theorem 5.2 (Corpus-grain incompleteness). *No self-reading of the formal corpus is universal for the predicates of its own level: names are outnumbered by predicates, and any completed self-portrait lags. Completeness-in-the-limit is a fair-schedule* policy, *not a theorem that the corpus catches itself.*

Remark 5.3 (Status). Theorems 5.1–5.2 are proved (Lawvere-style [13]; no arithmetisation). Host-logic incompleteness and Gödel coding of the corpus text in the sense of [10] are *refused* (Section 12).

6 Ladder

Dependence: *Theorem 5.1 (named escapes).*

Definition 6.1 (Naming extension and ladder). A *naming extension* adjoins a name for an unrepresented escape. The committed ladder sets level 0 to the fundamental carrier and each successor to an option type adjoining a namer for the previous dodge.

Theorem 6.2 (Ladder properties). *History embeds injectively and conservatively along the ladder; no level is exhausted by its predecessor; underdetermination persists at every level. Ascent is forced; choice of which escape is named is free.*

Remark 6.3 (Status). Proved in the foundation layer. Duration, as used below, means the order-type of this self-escape sequence.

Interpretation 6.4. The pattern "forced relation, free absolute"—forced ascent, free direction—reappears for channel labels in Section 7.

7 Registration: transferring losslessness

Spine acts are involutions. A ledger microstep is a forward map $U : S \times E \to S \times E$. The interface demand is that global injectivity of U import Proposition 3.4. Reading injectivity as "no erasure" follows the logical-reversibility tradition [1, 9]; its thermodynamic counterpart [12] is confined to the dictionary layer (Section 11).

Definition 7.1 (Two-bounce factorisation). A *two-bounce factorisation* of $U : \alpha \to \alpha$ is a pair of involutions (ϕ, σ) with $U = \sigma \circ \phi$ pointwise.

Theorem 7.2 (Converse). *If U admits a two-bounce factorisation, then U is lossless. On a product carrier this is ledger injectivity.*

Theorem 7.3 (Existence on finite carriers). *Every injective $U : \mathrm{Fin}\,n \to \mathrm{Fin}\,n$ admits a two-bounce factorisation. The construction is cycle-wise reflection at a least-value basepoint, with $\sigma := U \circ \phi$.*

Remark 7.4 (Status). Theorems 7.2–7.3 are proved. On finite carriers, injectivity and two-bounce form coincide: existence recovers, constructively, the classical fact that every finite permutation is a product of two involutions [16], and maps so factorised are the *reversible* maps of dynamical-systems theory [11]. Special cases (involutions as $U \circ \mathrm{id}$; Boolean injections; swap step) are included.

Theorem 7.5 (Registration shape)**.** *Under joint injectivity, a system merge writes a record in the environment (registration). Minimal merge alphabet size is* 2*, matching* $|\mathsf{Fund}|$.

Remark 7.6 (Status). Proved in the dynamics layer, using the injectivity interface above.

7.1 Canonicity of the factor pair

Theorem 7.7 (Canonicity fails)**.** *There exist an injective* $U : \mathsf{Bool} \to \mathsf{Bool}$ *and two two-bounce factorisations that disagree on* ϕ *(equivalently:* $\neg$ *factors as both* $\neg \circ \mathrm{id}$ *and* $\mathrm{id} \circ \neg$*). Moreover, on* $\mathrm{Fin}\, 3$ *two reflection-axis factorisations of a* 3*-cycle disagree and are not related by bounce-swap* $(\phi, \sigma) \mapsto (\sigma, \phi)$.

Remark 7.8 (Status). Proved negative. The absolute position of the factor pair is a *packaging* gauge (Section 10): the ledger consumes U, never (ϕ, σ).

Theorem 7.9 (Channel-swap is reversal)**.** *Let* $U = \sigma \circ \phi$ *with* ϕ, σ *involutions. Then* $\phi \circ \sigma$ *is a two-sided inverse of* U*, and* (σ, ϕ) *is a two-bounce presentation of that reverse step. The construction is independent of which factorisation of U was chosen.*

Proof sketch. From $U = \sigma \circ \phi$ and the involution equations, $\phi(\sigma(U(x))) = \phi(\sigma(\sigma(\phi(x)))) = \phi(\phi(x)) = x$ and $U(\phi(\sigma(x))) = \sigma(\phi(\phi(\sigma(x)))) = \sigma(\sigma(x)) = x$. The swapped pair (σ, ϕ) presents the reverse step. Uniqueness of (ϕ, σ) is not used. □

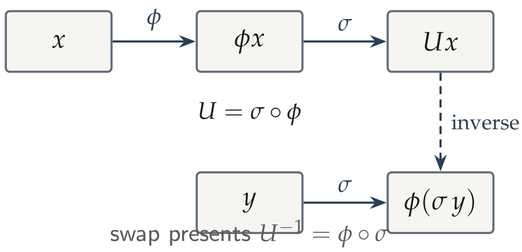


Figure 2: *Two-bounce presentation and swap. Pair labels are conventional; exchange covers reversal (Theorem 7.9).*

Interpretation 7.10. In the dihedral picture [6], a rotation fixes the angle between mirrors, not their absolute position. Theorem 7.7 is that fact on finite carriers; Theorem 7.9 promotes the exchange.

8 Archive and capacity

Dependence: *ladder carrier (Section 6) as environment; registration (Theorem 7.5).*

Theorem 8.1 (Archive and capacity)**.** *Taking the environment at horizon* T *to be the ladder carrier at level* T*, the committed capacity schedule satisfies* $|\mathrm{caps}\, T| = 2^{T+2}$*. Joint injectivity, merge, muting, and confinement imply the combinatorial bound* $K^T \leq |\mathrm{caps}\, T|$*; at* $K = 2$ *the committed ladder is eternally alive by arithmetic.*

Remark 8.2 (Status). Proved in the dynamics layer. Continuum area or continuum entropy identifications are refused.

9 Arrow as address

Dependence: *registration and archive growth.*

Theorem 9.1 (Arrow reading)**.** *The dynamical arrow—archive ascent under registration—is identified with address valued in* $\{\mathsf{fwd}, \mathsf{rev}\}$*. Tick identification (naming tick with microtick) holds for the classified fragment of uniform free archives up to record gauge; periodic fund-exempt dynamics and swap necessity are part of that classification.*

Remark 9.2 (Status). Proved as a classified package in the dynamics layer. Record gauge is packaging: exported only inside "up to record gauge" clauses.

10 One underdetermination and gauges

Dependence: *pole swap from Theorem 4.2; address from Theorem 9.1; channel from Definition 7.1 and Theorem 7.9.*

Theorem 10.1 (One $\mathbb{Z}/2\mathbb{Z}$, three faces)**.** *There is a single two-element underdetermination with three named involutive projections:*

$$\textit{pole swap} \quad \cong \quad \textit{address}\ \{\mathsf{fwd}, \mathsf{rev}\} \quad \cong \quad \textit{channel}\ \{\varphi, \sigma\}.$$

Each projection is an involution on a common $\mathbb{Z}/2\mathbb{Z}$ *carrier.*

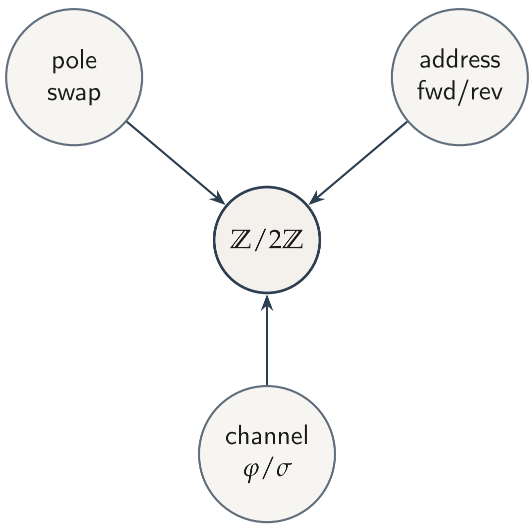


Figure 3: *Three faces of one underdetermination (Theorem 10.1). Faces are presentations; the group is stated once.*

Definition 10.2 (Content vs packaging)**.** A gauge is *content* when some exported statement mentions the choice. A gauge is *packaging* when the choice is quotiented before downstream consumption.

Table 1: *Gauge demarcation protecting the single dictionary $\mathbb{Z}/2\mathbb{Z}$.*

Gauge	Locus	Class
Label / pole swap	Theorem 4.2	content
Address fwd/rev	Theorem 9.1	content
Channel swap	Theorem 7.9	content
Record gauge per tick	Theorem 9.1	packaging
Factorisation gauge per step	Theorems 7.3–7.7	packaging

Remark 10.3 (Status). Theorem 10.1 and Definition 10.2 keep one dictionary $\mathbb{Z}/2\mathbb{Z}$: factorisation non-uniqueness (Theorem 7.7) is packaging; channel *labels* are conventional; channel *swap* is content. The content/packaging demarcation is a formal analogue of the question which symmetries carry representational significance [4, 8]; underdetermination by label swap is the formal shadow of permutation-style underdetermination of reference [18, 17].

11 Dictionary interpretations

Interpretation 11.1. Dictionary entries are optional certificates that a physical or contemplative model realises kernel symmetries (for example, swap as negation; dissipation as registration, in the sense of Landauer's principle [12]). They do not derive continuum spacetimes or observational numbers from Bool.

Remark 11.2 (Status). Layer 3 only. No theorem in Sections 3–10 depends on a dictionary certificate.

12 Open, refused, and philosophical

Continuum Lorentzian structure and boosts (no continuous one-parameter subgroup from a single $\mathbb{Z}/2\mathbb{Z}$); sheaf assembly beyond combinatorial stalk agreement; recombination budget versus ultrametric obstruction; alphabet-lift beyond the Fin interfaces closed here.

Premise-free global laws of time; continuum Einstein equations as theorems of this corpus; host-logic incompleteness; Gödel arithmetisation of the formal text; experiential or qualia claims; derivation of the world from Bool alone.

Facticity of liveness; adequacy of the pointed unoriented ambient (Interpretation 3.6); which face of the $\mathbb{Z}/2\mathbb{Z}$ is treated as free in applications.

13 Relation to prior work

Diagonal and fixed-point arguments. The seal (Theorem 5.1) instantiates Lawvere's diagonal schema [13], the common form of Cantor's theorem [5] and, via arithmetisation, of Gödel's first incompleteness theorem [10]; Yanofsky [19] surveys the family. Those applications fix a logical or set-theoretic universe and derive a limitative theorem inside it. Here the carrier of the diagonal argument is itself the output of the classification (Sections 3–4): the seal is a structural property of any fundamental act, not of a formal system. No arithmetisation is performed, and host-logic incompleteness is refused, not claimed (Section 12).

Reversible computation. Reading injectivity as "no erasure" follows the logical-reversibility tradition [1, 9], with Landauer's principle [12] as its thermodynamic counterpart. That literature takes reversibility as a premise about computing machinery. Here losslessness is not a premise: erasure is excluded in-arena by one equation (Proposition 3.4), and reversibility of forward microsteps is recovered by the two-bounce factorisation (Theorems 7.2–7.3). The thermodynamic reading survives only as a dictionary certificate (Section 11).

Involutions and reversing symmetries. That every finite permutation is a product of two involutions is classical [16], as is the dihedral factorisation of a rotation into reflections [6]; maps so factorised are the reversible maps of dynamical-systems theory [11]. The contribution here is not the existence of the factorisation but the disposition of its non-uniqueness: canonicity fails (Theorem 7.7), the failure is classified as packaging rather than content (Definition 10.2), and channel swap—which is content—covers step reversal independently of the chosen factorisation (Theorem 7.9).

Symmetry and underdetermination. Which symmetries carry representational significance is a standing question in the philosophy of physics [4, 8]; permutation-style underdetermination of reference descends from Quine [18] and Putnam's model-theoretic argument [17]. Those debates are conducted informally over rich physical or semantic theories. The analogue here is deliberately narrow: the content/packaging distinction is a definition in the formal text (Definition 10.2), each gauge is classified by theorem (Table 1), and the residual underdetermination is exactly inventoried—one $\mathbb{Z}/2\mathbb{Z}$, three faces (Theorem 10.1). Nothing is asserted about physical theories; within its scope the inventory is complete and machine-checked.

Machine-checked philosophical argument. Formalising a philosophical argument in a proof assistant has precedent, notably the automation of Gödel's ontological argument [2]. The role of verification differs here: the proof assistant certifies not a single argument but an architecture—downward-only imports across three layers (Section 2), axiom footprints audited per declaration, and interpretation confined to a layer the formal text cannot import.

14 Conclusion

The classification theorem restricts self-actions to the live two-cycle under the stated ambient. Fundamentality is the universal property of that survivor: the unique object is $(\mathsf{Bool}, \neg)$ up to label swap. Seal, ladder, registration (via two-bounce), archive capacity, and arrow as address follow in dependency order. One $\mathbb{Z}/2\mathbb{Z}$ is stated once; content and packaging gauges prevent factorisation non-uniqueness from counting as a second underdetermination. Philosophical and dictionary claims are confined to marked layers and to Section 12.

Formal companion

Representative declarations (Lean 4; no Mathlib; no `sorry`):

`classification_at` · `canon_initial` · `symmetric_excludes_erase` · `seal_on_fundamental` · `corpus_not_universal` · `one_Z2_three_faces` · `factor_fin` · `canonicity_fails` · `channel_swap_is_reversal` · `registration` · `caps_eq` · `tick_identification`

Axiom footprints are audited per declaration.

References

[1] C. H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.

[2] C. Benzmüller and B. Woltzenlogel Paleo. Automating Gödel's ontological proof of God's existence with higher-order automated theorem provers. In *ECAI 2014*, volume 263 of *Frontiers in Artificial Intelligence and Applications*, pages 93–98. IOS Press, 2014.

[3] G. Boole. *An Investigation of the Laws of Thought, on Which Are Founded the Mathematical Theories of Logic and Probabilities*. Walton and Maberly, London, 1854.

[4] K. Brading and E. Castellani, editors. *Symmetries in Physics: Philosophical Reflections*. Cambridge University Press, Cambridge, 2003.

[5] G. Cantor. Über eine elementare Frage der Mannigfaltigkeitslehre. *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 1:75–78, 1891.

[6] H. S. M. Coxeter. *Regular Polytopes*. Dover, New York, third edition, 1973. (Dihedral factorisation of rotations as products of reflections.)

[7] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction — CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021.

[8] N. Dewar. Sophistication about symmetries. *British Journal for the Philosophy of Science*, 70(2):485–521, 2019.

[9] E. Fredkin and T. Toffoli. Conservative logic. *International Journal of Theoretical Physics*, 21(3–4):219–253, 1982.

[10] K. Gödel. Über formal unentscheidbare Sätze der *Principia Mathematica* und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931.

[11] J. S. W. Lamb and J. A. G. Roberts. Time-reversal symmetry in dynamical systems: a survey. *Physica D*, 112(1–2):1–39, 1998.

[12] R. Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.

[13] F. W. Lawvere. Diagonal arguments and cartesian closed categories. In *Category Theory, Homology Theory and their Applications II*, volume 92 of *Lecture Notes in Mathematics*, pages 134–145. Springer, 1969. Reprinted with author commentary in *Reprints in Theory and Applications of Categories*, 15:1–13, 2006.

[14] S. Mac Lane. *Categories for the Working Mathematician*. Volume 5 of *Graduate Texts in Mathematics*. Springer, New York, second edition, 1998.

[15] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. ACM, 2020.

[16] T. K. Petersen and B. E. Tenner. How to write a permutation as a product of involutions (and why you might care). *Integers*, 13:#A63, 2013.

[17] H. Putnam. Models and reality. *Journal of Symbolic Logic*, 45(3):464–482, 1980.

[18] W. V. Quine. Ontological relativity. In *Ontological Relativity and Other Essays*, pages 26–68. Columbia University Press, New York, 1969.

[19] N. S. Yanofsky. A universal approach to self-referential paradoxes, incompleteness and fixed points. *Bulletin of Symbolic Logic*, 9(3):362–386, 2003.